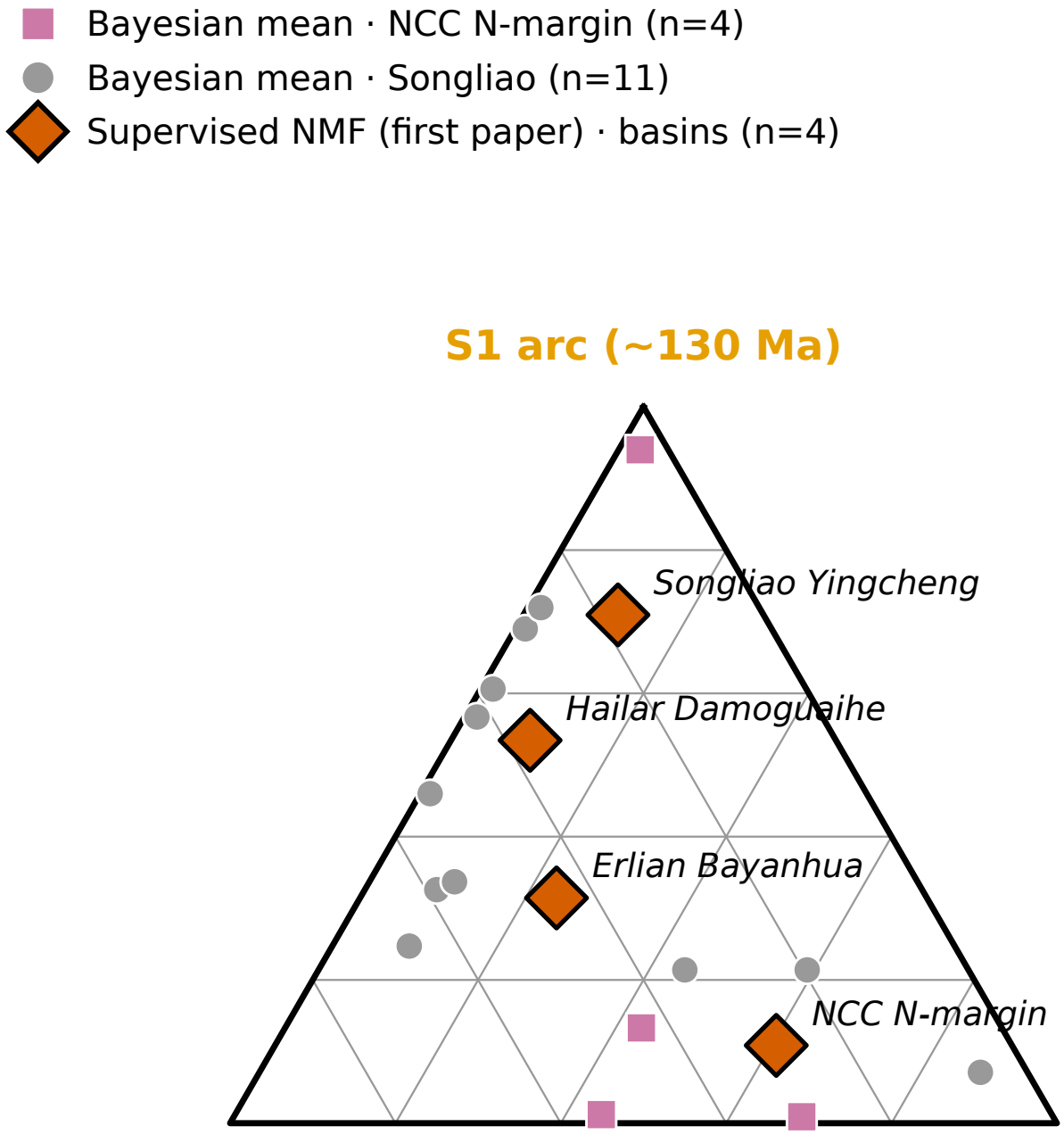


⚠ *Different sample sets: first-paper basins are literature-grounded syntheses, this study uses 15 real OneDZ/Puetz samples — provenance-pattern consistency, not per-sample validation.*